G H Raisoni International Skill Tech University, Pune

Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956

End Semester Examinations - 2025

School of Engineering and Technology

S.Y. B.Tech., Semester - III (CE/CS/AIDS/AIML/AI/Cyber)

Course Name: Computer Architecture & Organization **Course Code:25BTPCL303**

Max Marks: 60 **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D is compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	What are different functional units of computer organization in detail?	5	BL2	CO1
	B	Explain Different Addressing Modes in Details.	5	BL3	CO1
	C	Differentiate Between RISC & CISC with its Characteristics.	5	BL3	CO1
	D	Explain Types of Memory with suitable example.	2	BL2	CO1
2		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Explain IEEE 754 format standards for floating point representation with example.	5	BL2	CO2
	B	Explain Fixed & Floating Point algorithms with example.	5	BL2	CO2
	C	Explain Booths Algorithm with suitable example.	5	BL3	CO2
	D	Explain Restoring division algorithm with example.	2	BL2	CO2

3		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Draw and explain the Memory Hierarchy. What is the general relationship among access time, memory cost, and size?	5	BL2	CO3
	B	Explain any one type of Cache Mapping Technique with diagram.	5	BL2	CO3
	C	Exlain different Memrory Access Methods in Details.	5	BL3	CO3
	D	Define : CaChe Memory, Hit Ratio,Auxiliary Memory	2	BL2	CO3
4		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Define Pipelining ? Explain Key Characterstics of Piplining in detail.	5	BL2	CO4
	B	Discuss the varioustypes of Pipeline Hazards with example.	5	BL2	CO4
	C	Draw & EXPlain Single Data Path in Pipeling in detail.	5	BL2	CO4
	D	Explain different types of Exceptions in Pipelining in Details..	2	BL3	CO4
5		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Explain the different Control Unit Functions in Details.	5	BL2	CO5
	B	Compare & Contrast between Hardwired vs Micro-programmed control Unit .	5	BL2	CO5
	C	Draw and explain in detail block diagram of hardwired control unit with its Characterstics.	5	BL2	CO5
	D	Explain Advantages & Disadvantages of Micro Programmed Control Design.	2	BL2	CO5

- End of the Question Paper -